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The Polarization Paradox: How Suppressing Backfire Cascades Entrenches Ideological Segregation under Fragile Trust

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Abstract: Ideological polarization on social media can harm both public discourse and platform economics, notably through user exit and lower advertising value. Platforms often respond with bridging-style feed interventions that reduce exposure to confrontational, cross-cutting content in what Social Judgment Theory calls the latitude of rejection. We study the long-term effects of such interventions with Dynamic-BABE (Biased Assimilation & Behavioral Entrenchment), an agent-based model that combines biased assimilation, social-judgment zones, and co-evolving dyadic trust. A 2×2 design crosses bridging (ON/OFF) with trust co-evolution (ON/OFF) over 30 replications per cell ($N = 500$, 200 steps). We find a Polarization Paradox: bridging cuts cumulative churn from 39.18% to 0.38% ($d = 23.97$, $p < 0.001$) and raises step-wise revenue from \$121.64 to \$199.24 ($d = -23.97$, $p < 0.001$), yet final polarization rises from 0.9550 to 0.9904 ($d = -1.28$, $p < 0.001$). Separately, enabling trust alone (Trust Only vs. Baseline) produces strong trust segregation (35.61 ± 31.50; median $= 28.58$; $d = -1.55$, $p < 0.001$). Seed-paired Bridge $\times$ Trust tests show that retention gains remain robust when trust is active, while relational moderation is modest. In short, interventions that reduce exit need not improve cohesion when disagreement is suppressed and trust is fragile.

Keywords: agent-based modeling; opinion dynamics; polarization; dyadic trust; algorithmic curation; social media

Introduction

1.1 Digital platforms now structure much of the public sphere. They also concentrate disagreement: algorithmic feeds can amplify conflict and deepen affective polarization (Bail et al. 2018; Levy 2021).

1.2 For platforms, polarization is an economic problem as well as a social one. Hostile feeds raise cognitive load; users may leave. Hirschman (1970) called this exit. Exit shrinks the active audience and advertising impressions. Advertisers may also discount unsafe feeds. In this paper the “brand safety penalty” is a stylized revenue assumption, not a calibrated estimate from a named firm. Empirical work on feed ranking and attitudes (Guess et al. 2023; Levy 2021) motivates attention to curation, but does not measure that multiplier directly.

1.3 Platforms therefore experiment with structural feed interventions that reduce exposure to confrontational content before conflict escalates. We model one such intervention as a probabilistic “bridging” filter. It intercepts interactions in the latitude of rejection (Sherif & Hovland 1961)—messages so far from the receiver's position that exposure tends to trigger backlash rather than assimilation. Field experiments (Guess et al. 2023; Bail et al. 2018) show that exposure regimes shape attitudes and behavior; our bridge is an abstraction of suppression-oriented curation, not a claim about any specific product.

1.4 Suppression can create a trade-off we call the Polarization Paradox. Bridging may reduce churn, frustration, and revenue loss in the short run while failing to reduce—and sometimes increasing—systemic polarization. By removing cross-cutting friction, the platform can also remove opportunities to rebuild tolerance.

1.5 We formalize this with Dynamic-BABE (Biased Assimilation & Behavioral Entrenchment). The model extends classical opinion dynamics (DeGroot 1974; Hegselmann & Krause 2002) by combining entrenchment-based biased assimilation (Chen et al. 2021) with Social Judgment Theory (Sherif & Hovland 1961; Jager & Amblard 2005) and co-evolving interpersonal trust. Following Slovic (1993), trust builds slowly under agreement and erodes quickly under conflict.

1.6 A 2×2 experiment crosses bridging with dynamic trust. Bridging dampens immediate conflict and exit, but can leave extreme agents active who would otherwise leave. Separately, trust co-evolution produces trust segregation: trust concentrates within opinion camps while outgroup trust collapses. That pattern is distinct from the bridging paradox.

1.7 Our contributions are threefold. First, Dynamic-BABE links cognition, dyadic trust, and stylized platform economics in one ABM. Second, we document the Polarization Paradox under suppression-oriented bridging. Third, we argue that retention metrics alone are incomplete if the goal includes relational cohesion.

1.8 Replication materials are available at `https://github.com/TheCoolSam/BABEModel`; we intend to deposit the model on CoMSES Net for permanent archival.[1]

Related Work

2.1 We situate Dynamic-BABE in five literatures: classical opinion dynamics, biased assimilation, Social Judgment Theory, co-evolving trust, and algorithmic exposure.

Classical Models of Opinion Dynamics

2.2 DeGroot (1974) modelled influence as repeated weighted averaging. In a connected network this process converges to consensus, which cannot explain persistent disagreement. Bounded-confidence models restrict influence to neighbours within a similarity threshold (Hegselmann & Krause 2002; Deffuant et al. 2000) and can produce multiple clusters. Flache et al. (2017) review these and related formalisms. Bounded confidence typically ignores out-of-range neighbours rather than modelling active repulsion.

Cognitive Resistance and Biased Assimilation

2.3 Lord et al. (1979) showed that people can interpret mixed evidence in ways that strengthen prior attitudes—biased assimilation. Dandekar et al. (2013) embedded related biases in network models and linked them to polarization under homophily. Chen et al. (2021) proposed BEBA, where influence weights $w_{ij}(t) = 1 + \beta_i y_i(t) y_j(t)$ depend on entrenchment β_i and alignment, so negative influence can emerge endogenously. Mäs & Flache (2013) discuss bipolarization without explicit distancing. We adapt Chen-style weights to pairwise, multi-issue interactions, map them onto Social Judgment zones (Sherif & Hovland 1961; Jager & Amblard 2005), and add co-evolving dyadic trust.

Social Judgment Theory and Backfire

2.4 Social Judgment Theory partitions attitude space into latitudes of acceptance, non-commitment, and rejection (Sherif & Hovland 1961). Messages in the acceptance latitude tend to pull opinions toward the source; messages in the rejection latitude can produce contrast and backlash. Jager & Amblard (2005) showed that repulsion in networks can create opposing camps. We treat influence weights as determining these zones.

Networks and Dyadic Trust

2.5 Opinions and social structure often co-evolve (Macy & Willer 2002; Centola 2010; Keijzer et al. 2018). Many models allow rewiring toward similar others. We hold the undirected graph fixed and instead let edge trust change with interaction. Trust is asymmetric in gain and loss (Slovic 1993): it rises slowly after agreement and falls faster after conflict. High trust buffers occasional disagreement; low trust makes further conflict more likely.

Algorithmic Interventions and Exposure Experiments

2.6 Platforms once hoped that cross-cutting exposure would reduce polarization. Bail et al. (2018) found that exposing Twitter users to opposing-party bots could increase extremity. Feed experiments also show that ranking regimes change on-platform behavior, often more than deep attitudes (Levy 2021; Guess et al. 2023). We abstract a suppression-oriented regime as a probabilistic bridge that intercepts rejection-zone events. Field studies rarely observe long-run feedback among opinions, trust, and exit; the ABM is built for that gap.

Model Description (ODD+D Protocol)

3.1 We describe Dynamic-BABE with the ODD+D protocol (Grimm et al. 2010; Müller et al. 2013) so that entities, processes, and decision rules are explicit and reproducible.

Overview

Purpose

3.2 Dynamic-BABE is a theoretical laboratory for feedback among interpersonal trust, biased assimilation / backfire, and stylized feed bridging. We use it to study the Polarization Paradox: bridging can cut exit and protect revenue while leaving (or raising) systemic polarization.

Entities, State Variables, and Scales

3.3 Two entities drive the simulation: agents (users navigating their feeds) and the network environment (the network's structural pathways and the platform's moderation layer).

Agents:

3.4 Each agent $i \in \{1, 2, \ldots, N_{\text{agents}}\}$ represents a single user, tracked via several state variables:

- Opinion Vector ($\vec{O}_i \in [-1, 1]^K$): A multi-dimensional ideological signature representing the agent's stance on K distinct social or political issues. The issue dimension $d \in \{1, \ldots, K\}$ spans continuously from -1.0 (progressive/left) to $+1.0$ (conservative/right).
- Salience Vector ($\vec{S}_i \in [0, 1]^K$): An attention allocation vector mapping the relative cognitive importance the agent assigns to each issue. We constrain this vector so that salience sums to unity: $\sum_{d=1}^{K} S_{i,d} = 1.0$.
- Cognitive Entrenchment ($\beta_i \in [\beta_{\min}, \infty)$): A scalar tracking cognitive resistance, dogmatism, or confirmation bias (Chen et al. 2021). Elevated β_i values amplify outgroup rejection and fuel stronger backfire repulsion.
- Behavioral Frustration ($F_i \in \mathbb{N}_0$): An integer counter tracking the psychological irritation and cognitive dissonance that mount during confrontational, cross-cutting exchanges.
- Active Status ($\text{Active}_i \in \{\text{True}, \text{False}\}$): A boolean flag marking platform activity. When frustration breaches the critical threshold T_c, the agent permanently exits the platform ($\text{Active}_i = \text{False}$), registering as user churn.

Network and Platform Environment:

3.5 We model the social topology as an undirected, scale-free graph $G = (V, E)$ generated via the Barabási-Albert (BA) preferential attachment algorithm (Barabási & Albert 1999). The node set V represents agents, while the edge set E maps communication pathways.

- Dyadic Interpersonal Trust ($T_{ij} \in [0, 1]$): A symmetric weight representing relational capital on the edge $(i, j) \in E$. Trust cushions disagreement and co-evolves with opinion alignment. Disabling the trust dynamic locks T_{ij} at a static, neutral baseline $T_0 = 0.5$ across all edges.
- Algorithmic Interception Layer: The platform's computational gatekeeper. When active, it intercepts impending backfire events, neutralizing the psychological sting of deep ideological disagreement.

Scales:

3.6 Time ticks in discrete steps $t \in \{1, 2, \ldots, t_{\max}\}$, where each step represents a conversational epoch. Space is purely topological, dictated by network connections. The model populates a scale-free network with $N_{\text{agents}} = 500$ agents and preferential attachment parameter $m_e = 3$, yielding degree distributions that mirror real-world social networks.

Process Overview and Scheduling

3.7 Each step t triggers a sequence of operations that update opinions, frustration, trust, and churn:

1. Randomized Activation Order: The scheduler shuffles all active agents into a randomized list at the start of each step to eliminate sequential bias.
2. Agent Interaction Loop: We iterate through each active agent i in the shuffled list:
   (a) The agent filters its neighbors to isolate active peers: $\mathcal{N}_i^{\text{active}}(t) = \{j \in \mathcal{N}_i : \text{Active}_j(t) = \text{True}\}$. If this neighborhood is empty, the agent skips its turn.
   (b) The agent selects a dialogue partner j from $\mathcal{N}_i^{\text{active}}(t)$ uniformly at random. We set the interaction step size to $k_{\text{step}} = 1$ to capture micro-conversational dynamics.
   (c) The directed interaction pipeline $\text{Interact}(i, j)$ executes, updating agent i's opinion vector, accumulating frustration, checking for churn, and updating the symmetric trust T_{ij} on the connecting edge.
3. Passive Trust Decay: Once all active interactions conclude, we apply passive decay to every edge $(u, v) \in E$, including edges that were updated during the same step:
$$T_{uv}(t+1) = T_{uv}(t) \cdot (1 - \lambda) \tag{1}$$
4. Platform Diagnostics: The model computes and logs global metrics: opinion polarization $\sigma(t)$, cumulative churn, user frustration, active user count, ingroup/outgroup trust divergence, and opinion-weighted clustering.
5. Termination Check: The simulation halts when the step count reaches $t_{\max} = 200$, or if all users exit ($N_{\text{active}} = 0$).

Design Concepts

Theoretical Background

3.8 We anchor Dynamic-BABE's cognitive architecture in Social Judgment Theory (Sherif & Hovland 1961; Jager & Amblard 2005). This framework partitions an agent's attitude spectrum into three distinct zones: the Latitude of Acceptance (yielding assimilative agreement), the Latitude of Non-Commitment (resulting in inertia), and the Latitude of Rejection (triggering backfire repulsion).

3.9 To formalize cognitive entrenchment (β_i) and opinion alignment (A_{ij}), we adapt the Biased Assimilation and Behavioral Entrenchment (BEBA) model proposed by Chen et al. (2021), which extends DeGroot's classical consensus dynamics (DeGroot 1974). We merge this with a dynamic trust layer inspired by Slovic's empirical research (Slovic 1993). This integration captures the fundamental asymmetry of trust: relational capital accumulates painfully slowly through agreement but collapses instantly during conflict.

3.10 Finally, we operationalize user exit by mapping the accumulation of frustration to the classic Exit-Voice-Loyalty (EVL) framework of Hirschman (1970).

Individual Decision-Making

3.11 Agents employ heuristic cognitive updates rather than rational utility maximization, strictly following the bounded rationality principles of the ODD+D protocol.

- Objectives: Agents do not maximize an explicit utility or payoff function. Instead, their immediate psychological objective is to minimize cognitive dissonance. They seek agreeable social contact that reinforces their existing views while avoiding highly confrontational outgroup viewpoints that land in their Latitude of Rejection.
- Information Processing: When active, an agent perceives the opinion vector $\vec{O}_j$ and salience weights $\vec{S}_j$ of their chosen dialogue partner. They process this information heuristically by calculating their salience-weighted opinion alignment A_{ij} and checking whether the resulting influence weight lands in their Latitude of Acceptance, Non-Commitment, or Rejection.
- Adaptive Behavior: Agents adapt dynamically across two scales. Cognitively, they adjust their opinion vectors $\vec{O}_i$ to assimilate toward agreeable peers or repel away from hostile ones. Behaviorally, they accumulate frustration F_i when clashing with others, leading to permanent exit (churn) if their frustration threshold is breached. Relationally, their dyadic trust T_{ij} adapts to reward agreement and penalize conflict, reshaping future interaction weights.

Learning

3.12 Although agents lack formal reinforcement learning capabilities, they maintain a relational memory via dyadic trust (T_{ij}). They continuously update their assessment of neighbor compatibility. Agreeable, assimilative dialogue slowly accumulates trust, whereas hostile backfires instantly incinerate it. This co-evolving relational memory shapes future influence weights, closing the feedback loop between network relations and cognitive updates.

Individual Sensing

3.13 Each agent i tracks its direct topological neighbors with perfect accuracy. During dialogue, agent i observes a neighbor's opinion vector $\vec{O}_j$ and salience weights $\vec{S}_j$. Agents remain blind to the state variables, cognitive profiles, and degrees of non-neighboring nodes, and they hold zero global knowledge regarding system-level polarization or platform-level financials.

Individual Prediction

3.14 Agents react purely to immediate events, operating without forward-looking predictions. They neither anticipate neighbor opinion shifts nor forecast their own impending exit as frustration mounts toward the churn threshold.

Interaction

3.15 Interactions unfold locally, dyadically, and directionally. While trust updates symmetrically across the edge (i, j), opinion and frustration updates execute from the perspective of the active agent receiving the communication. The platform's bridging algorithm acts as an external filter, intercepting high-friction exchanges to suppress the behavioral symptoms of disagreement.

Collectives

3.16 No formal collectives (such as political parties or coalitions) exist in our model. However, functional collectives emerge endogenously through conversational dynamics. Agents coalesce into two opposing ideological factions based on their opinion signs. These emergent camps govern the flow of trust and drive the self-organization of network clusters.

Heterogeneity

3.17 We introduce substantial heterogeneity across several parameters:

- Ideological Camps: Bipolar initial opinions split the population into two opposed factions.
- Issue Salience: Diverse attention allocations drawn from a Dirichlet distribution mean agents prioritize issues differently.
- Cognitive Entrenchment: The entrenchment parameter β_i varies across agents, mixing dogmatic and open-minded individuals.
- Network Topology: The Barabási-Albert topology yields a heavy-tailed degree distribution where a few highly connected hubs exert massive social influence while the rest have minimal reach.

Stochasticity

3.18 We integrate stochasticity to capture social noise and scheduling dynamics:

- The preferential attachment model generates network topologies stochastically.
- Shuffling shatters sequential order by randomizing agent activation at each step.
- Dialogue partner selection is stochastic, choosing randomly from active neighbors.
- The bridging algorithm operates probabilistically, intercepting rejection events with an efficacy rate of $E = 46\%$.
- Initial values for opinion, salience, and entrenchment are drawn randomly from their respective probability distributions.

Observation

3.19 We record the system's trajectory at the end of each step by collecting system-level and agent-level diagnostics:

- Global Polarization ($\sigma(t)$): The root-mean-square (RMS) of per-dimension opinion standard deviations across active agents.
- Active User Count (N_{active}): The sum of active nodes.
- Churn Rate: The cumulative fraction of the population that has permanently exited.
- Average Frustration: The mean frustration across active users.
- Platform Revenue ($R(t)$): Total advertising earnings, which drop when polarization triggers advertiser flight.
- Trust Segregation Ratio: The ratio of mean ingroup trust to mean outgroup trust.
- Opinion-Weighted Clustering: The density of triadic relationships where all three connected nodes share the same opinion pole.

Details

3.20 This section details the mathematical formulations and structural rules governing initialization, dialogue, trust co-evolution, user exit, and platform monetization.

Initialization

3.21 At $t = 0$, the model builds an undirected, scale-free network G of $N_{\text{agents}} = 500$ nodes with attachment parameter $m_e = 3$.

3.22 To seed a pre-polarized society, we initialize opinion vectors $\vec{O}_i \in [-1,1]^K$ ($K = 2$ issues) via a bipolar split. Each agent is assigned a camp pole $\pi_i \in \{-1,+1\}$ with equal probability. Independently for each issue dimension d, extremity is drawn from a uniform interval and scaled by the pole:

$$O_{i,d}(0) = \Pi_{[-1,1]}\Big(\pi_i \cdot U_{i,d}(0.4, 1.0)\Big) \tag{2}$$

where $U_{i,d}(0.4, 1.0)$ denotes an independent draw from $\mathcal{U}(0.4, 1.0)$ and $\Pi_{[-1,1]}$ clips opinions to $[-1, 1]$. The expected absolute extremity is 0.7, yielding two opposed, pre-polarized factions without additional Gaussian noise.

3.23 We draw each agent's salience vector $\vec{S}_i$ from a symmetric Dirichlet distribution:

$$\vec{S}_i \sim \text{Dirichlet}(\vec{\alpha}) \tag{3}$$

where $\vec{\alpha} = [1, 1]^T$ for $K = 2$ issues, ensuring that $S_{i,d} \in [0, 1]$ and $\sum_{d=1}^{K} S_{i,d} = 1.0$.

3.24 To model cognitive dogmatism, we draw agent entrenchment β_i from a normal distribution and floor it to prevent negative resistance:

$$\beta_i = \max\Big(\beta_{\min},\, \nu_i\Big), \quad \nu_i \sim \mathcal{N}(\mu_\beta,\, \sigma_\beta^2) \tag{4}$$

with parameters set to default values: $\beta_{\min} = 0.5$, $\mu_\beta = 4.0$, and $\sigma_\beta = 1.5$.

3.25 All active network edges $(u, v) \in E$ begin with symmetric dyadic trust set to a uniform starting value:

$$T_{uv}(0) = T_0 = 0.5 \tag{5}$$

Every agent begins the simulation with zero behavioral frustration ($F_i(0) = 0$).

Input Data

3.26 The Dynamic-BABE model is a theoretical, self-contained simulation and does not ingest external empirical input data for its execution.

Submodels

Dialogue Dynamics and Opinion Updates:

3.27 Upon activation, agent i randomly selects an active neighbor j for dialogue, processing the incoming opinion through a multi-stage cognitive pipeline.

Step 1: Salience-Weighted Opinion Alignment (A_{ij})

3.28 We calculate the ideological distance between receiver i and sender j as a salience-weighted dot product:

$$A_{ij} = \sum_{d=1}^{K} O_{i,d} \cdot \left(\frac{S_{i,d} + S_{j,d}}{2}\right) \cdot O_{j,d} \tag{6}$$

This formulation captures a key psychological reality: mutually important issues—where salience is high—contribute disproportionately to perceived agreement or disagreement. Because opinions are bounded in

$[-1, 1]$ and average saliences sum to 1.0, the alignment spans $A_{ij} \in [-1, 1]$, where +1.0 represents perfect alignment on highly valued topics and −1.0 marks absolute ideological clash.

Step 2: Perceived Influence Weight (w_{ij})

3.29 Adapting the BEBA model (Chen et al. 2021), we establish a baseline influence weight based on the receiver's cognitive entrenchment and opinion alignment:

$$w_{ij}^{\text{base}} = 1.0 + \beta_i A_{ij} \tag{7}$$

Under the Dyadic Trust system, relational trust modulates this weight, serving as a positive cognitive buffer:

$$w_{ij} = 1.0 + \beta_i A_{ij} + \gamma_T T_{ij} \tag{8}$$

where $\gamma_T \geq 0$ represents the trust influence coefficient ($\texttt{TRUST\_INFLUENCE} = 0.3$) and $T_{ij} \in [0, 1]$ represents the dyadic trust. This trust buffer mathematically models the relational buffer that keeps relationships intact during disagreement: high trust elevates the influence weight, preventing debate from deteriorating into hostile backfires. When trust co-evolution is disabled, the model sets $T_{ij} = 0$ dynamically in the weight equation, yielding the baseline form $w_{ij} = 1.0 + \beta_i A_{ij}$. This represents the unmodulated, baseline cognitive state and aligns the theoretical specification with our simulation codebase.

Step 3: Social Judgment Zone Classification

3.30 Unlike the collective, synchronous averaging of Chen et al. (Chen et al. 2021), we translate this cognitive logic into a local, pairwise interaction paradigm. To track individual metrics like frustration, churn, and algorithmic interventions, we classify interactions using Social Judgment Theory (Sherif & Hovland 1961; Jager & Amblard 2005). The perceived influence weight w_{ij} dictates the interaction zone:

$$\text{Zone}(w_{ij}) = \begin{cases} 1 & \text{(Latitude of Acceptance)} & \text{if } w_{ij} > \tau_{\text{assim}} \\ 2 & \text{(Latitude of Non-Commitment)} & \text{if } 0.0 \leq w_{ij} \leq \tau_{\text{assim}} \\ 3 & \text{(Latitude of Rejection)} & \text{if } w_{ij} < 0.0 \end{cases} \tag{9}$$

where $\tau_{\text{assim}} = 0.3$ represents the assimilation threshold. We model these latitude boundaries as stylized, uniform thresholds across the population for mathematical tractability and to isolate the systemic co-evolutionary feedback loops. In empirical social systems, these latitudes are highly subjective and dynamic, varying widely across individuals based on issue-specific emotional involvement and prior ideological extremity. We discuss the implications of this stylized assumption in Section .

Step 4: Algorithmic Moderation (The Bridge Algorithm)

3.1 When active, the platform's bridging algorithm intercepts backfire (Zone 3) events with an efficacy rate of $E = 0.46$:

$$w_{ij} \leftarrow 0.0, \quad \text{Zone}(w_{ij}) \leftarrow 2 \quad \text{with probability } E \tag{10}$$

Interception neutralizes the opinion repulsion, downgrading the conflict to Zone 2 (Ignore). In addition, successful moderation grants a frustration relief bonus to preserve the agent's psychological state. We model this frustration relief bonus ($B_{hb} = 2$) as the platform's active delivery of "dopaminergic comfort" or "algorithmic reinforcement" (e.g., stochastically substituting a hostile outgroup post with a soothing, highly engaging ingroup post or distracting entertainment). This intervention actively diverts attention, helping to soothe and heal preexisting cognitive friction.

Step 5: Opinion Update Rules

3.2 We update the opinion vector $\vec{O}_i$ depending on the final interaction zone:

- Zone 1 (Assimilation): The agent assimilates toward the sender using a DeGroot-style consensus update:

$$\vec{O}_i(t+1) = \Pi_{[-1,1]} \left(\vec{O}_i(t) + \mu_{ij} \left(\vec{O}_j(t) - \vec{O}_i(t) \right) \right) \tag{11}$$

where we attenuate the step size μ_{ij} by the agent's cognitive entrenchment:

$$\mu_{ij} = \frac{w_{ij}}{1.0 + \beta_i} \tag{12}$$

This ensures that highly entrenched agents (high β_i) make smaller opinion adjustments, reflecting cognitive resistance. The vector-wise operator $\Pi_{[-1,1]}$ acts element-wise to clip (clamp) updated opinions to the valid range $[-1.0, 1.0]$. We note that under conditions of absolute alignment ($A_{ij} = 1.0$) and high dyadic trust ($T_{ij} = 1.0$), the influence step size can stochastically exceed unity ($\mu_{ij} > 1.0$). While classical DeGroot consensus models enforce a strict convex combination ($\mu_{ij} \in [0, 1]$) to maintain the convex hull of opinions, our formulation permits this mathematical "overshoot." In the context of relational network dynamics, this represents a "passionate consensus" or "social reinforcement effect"—a psychological phenomenon where deep interpersonal trust and absolute agreement trigger cognitive enthusiasm, prompting the receiver to adopt a stance even more extreme than their trusted peer.

- Zone 2 (Non-Commitment): The interaction is ignored, leaving the opinion vector unchanged:

$$\vec{O}_i(t+1) = \vec{O}_i(t) \tag{13}$$

- Zone 3 (Backfire): The agent experiences ideological repulsion and pushes its opinion vector away from the sender:

$$\vec{O}_i(t+1) = \Pi_{[-1,1]}\left(\vec{O}_i(t) + \text{repulsion}_{ij}\left(\vec{O}_i(t) - \vec{O}_j(t)\right)\right) \tag{14}$$

where the repulsion step size is:

$$\text{repulsion}_{ij} = \frac{|w_{ij}|}{1.0 + \beta_i} \tag{15}$$

This repulsive force drives agent i further toward the ideological extremes, modeling the backfire effect, subject to the same element-wise clipping operator $\Pi_{[-1,1]}$.

Behavioral Frustration and User Churn:

3.3 Unmitigated disagreements accumulate psychological costs, which we model as behavioral frustration F_i:

- Frustration Accrual (Backfire): Unmitigated backfires (Zone 3) increment frustration:

$$F_i(t+1) = F_i(t) + 1 \tag{16}$$

- Frustration Healing (Assimilation): Assimilative dialogue (Zone 1) relieves cognitive dissonance, healing frustration at a constant rate $H = 1$:

$$F_i(t+1) = \max\left(0,\ F_i(t) - H\right) \tag{17}$$

- Moderation Relief Bonus: Successful algorithmic interception spares the agent the backfire and grants a mental relief bonus $B_{hb} = 2$:

$$F_i(t+1) = \max\left(0,\ F_i(t) - B_{hb}\right) \tag{18}$$

3.4 At the end of each interaction, the agent checks its psychological tolerance. When cumulative frustration breaches the churn threshold $T_c = 15$, the user permanently exits the platform ($\text{Active}_i = \text{False}$):

$$\text{Active}_i(t+1) = \begin{cases} \text{False} & \text{if } F_i(t+1) > T_c \\ \text{Active}_i(t) & \text{otherwise} \end{cases} \tag{19}$$

Churned users freeze instantly. They cease all dialogue, drop out of the communication graph, and generate zero advertising revenue.

Dyadic Trust Co-evolution:

3.5 Under the dynamic condition, dyadic trust T_{ij} co-evolves with dialogue outcomes:

$$T_{ij}(t+1) = \begin{cases} \min\Big(1.0,\, T_{ij}(t) + \delta_+\Big) & \text{if Zone}(w_{ij}) = 1 \text{ (Assimilation)} \\ \max\Big(0.0,\, T_{ij}(t) - \delta_-\Big) & \text{if Zone}(w_{ij}) = 3 \text{ (Backfire)} \\ T_{ij}(t) & \text{if Zone}(w_{ij}) = 2 \text{ (Ignore)} \end{cases} \tag{20}$$

where $\delta_+ = 0.02$ represents the slow accumulation of trust during agreement, and $\delta_- = 0.05$ captures the rapid collapse triggered by conflict. The resulting asymmetry ratio ($\delta_-/\delta_+ = 2.5$) mathematically formalizes Slovic's empirical trust asymmetry principle: social relationships are highly fragile, and conflict destroys trust much faster than cooperation builds it.

3.6 Because social ties naturally wither without active engagement, we apply a passive decay filter to every edge $(u, v) \in E$ at the end of each step:

$$T_{uv}(t+1) = T_{uv}(t) \cdot (1 - \lambda) \tag{21}$$

where $\lambda = 0.001$ represents the passive erosion rate.

Platform Revenue:

3.7 The social media platform monetizes active users, generating ad revenue at each step t. Advertisers flee volatile feeds, imposing a heavy Brand Safety Penalty if global polarization crosses the critical cliff:

$$R(t) = N_{\text{active}}(t) \cdot r_{\text{ad}} \cdot M_{\text{safety}}(t) \tag{22}$$

Here, $N_{\text{active}}(t)$ represents the active user count, $r_{\text{ad}} = \$1.00$ is the base ad rate per user, and $M_{\text{safety}}(t)$ represents the brand safety multiplier:

$$M_{\text{safety}}(t) = \begin{cases} M_{\text{safe}} = 1.0 & \text{if } \sigma(t) < \tau_{\text{cliff}} \\ M_{\text{unsafe}} = 0.4 & \text{if } \sigma(t) \geq \tau_{\text{cliff}} \end{cases} \tag{23}$$

where $\tau_{\text{cliff}} = 0.6$ represents the polarization cliff threshold. We calculate global polarization $\sigma(t)$ as the root-mean-square (RMS) of per-dimension standard deviations across active agents:

$$\sigma(t) = \sqrt{\frac{1}{K} \sum_{d=1}^{K} \text{SD}\Big(\{O_{i,d}(t) : i \in \mathcal{A}(t)\}\Big)^2} \tag{24}$$

where $\mathcal{A}(t) = \{i \in V : \text{Active}_i(t) = \text{True}\}$ and SD denotes the population standard deviation within dimension d. This formulation avoids the "data flattening" trap: averaging opinions across dimensions can cancel opposed extremes (e.g., $[+1, -1]$), whereas per-dimension dispersion preserved under RMS does not.

3.8 Ingroup and outgroup trust metrics classify an agent's pole by $\text{sign}(\overline{O}_i)$, the sign of the mean opinion across issues. Agents with mean opinion exactly zero ($\text{sign}(0) = 0$) are rare under bipolar initialization but, when present, are treated as a distinct zero pole for edge classification; we discuss this edge case as a limitation.

Experimental Design

3.9 We map the processes above with a 2×2 design crossing the Bridging Algorithm and dyadic trust:

1. Baseline (Bridge OFF, Trust OFF): Biased assimilation under static neutral trust ($T_{ij} = 0.5$).
2. Bridge Only (Bridge ON, Trust OFF): Bridging on; trust fixed.
3. Trust Only (Bridge OFF, Trust ON): Trust co-evolves; no bridging.
4. Full Model (Bridge ON, Trust ON): Both mechanisms active.

Table 1: Model Parameters and Default Values

Parameter Symbol	Description	Default Value	Source / Rationale
N_{agents}	Population size	500	Config.py standard pop. size
m_e	BA network attachment parameter	3	Scale-free social degree distribution
K	Number of opinion dimensions	2	Multidimensional issue space
β_{mean}	Mean of cognitive entrenchment	4.0	High entrenchment social media bias
β_{std}	SD of cognitive entrenchment	1.5	Cognitive heterogeneity in population
β_{min}	Minimum entrenchment floor	0.5	Prevent negative cognitive resistance
k_{step}	Dialogue partners selected per step	1	Micro-interaction conversational model
τ_{assim}	Assimilation threshold	0.3	Social Judgment Theory threshold
T_c	Churn frustration threshold	15	Toxic churn tolerance capacity
H	Natural frustration healing rate	1	Cognitive cooling per positive contact
B_{hb}	Bridge moderation healing bonus	2	Active relief under algorithmic safety
r_{ad}	Base advertising rate per step	\$1.00	Baseline monetization rate
τ_{cliff}	Brand safety polarization cliff	0.6	Advertiser flight tipping point
M_{safe}	Brand safe revenue multiplier	1.0	Normal ad monetization
M_{unsafe}	Brand unsafe revenue multiplier	0.4	60% discount due to toxic environment
E	Bridging algorithm efficacy	0.46	46% moderation success probability
T_0	Initial dyadic trust weight	0.5	Relational neutrality starting state
γ_T	Trust buffer influence coefficient	0.3	Disagreement buffering coefficient
δ_+	Trust building step size (gain)	0.02	Slow relational capital accumulation
δ_-	Trust erosion step size (loss)	0.05	Slovic relational fragility parameter
λ	Passive trust decay rate	0.001	Under-maintenance erosion rate
t_{max}	Maximum simulation step limit	200	Conversational equilibrium epoch

Model Hyperparameters and Topology

3.10 The model places $N = 500$ agents on a Barabási–Albert network (Barabási & Albert 1999) with $m = 3$. Opinions live in $K = 2$ issues with bipolar initialization: pole $\pi_i \in \{-1, +1\}$ with equal probability, and each issue set to $\pi_i \cdot U(0.4, 1.0)$ (clipped to $[-1, 1]$).

3.11 Entrenchment $\beta_i \sim \mathcal{N}(4.0, 1.5)$ with floor 0.5. When trust is on, influence shifts by $\gamma_T = 0.3$, gains $\delta_+ = 0.02$ on assimilation, loses $\delta_- = 0.05$ on backfire, and decays at $\lambda = 0.001$ per step. Bridging intercepts rejection-zone events with efficacy $E = 0.46$ and applies a two-point frustration relief.

3.12 We run $R = 30$ replications per condition for $S = 200$ steps. Run i uses seed $42 + i$.

Stylized Facts and Qualitative Targets

3.13 Parameters are literature-motivated, not fitted to a proprietary platform log. We instead check that default Dynamic-BABE recovers directional patterns reported in related work:

Backfire under opposing exposure. Rejection-zone contact moves opinions apart (Bail et al. 2018; Sherif & Hovland 1961).

Feed regime vs. deep attitudes. Bridging changes exit/retention strongly while polarization stays high or rises, consistent in spirit with feed experiments where ranking shifts behavior more than core attitudes (Guess et al. 2023).

Fragile trust. Gain/loss asymmetry ($\delta_-/\delta_+ = 2.5$) follows Slovic (1993) and yields trust segregation when trust is enabled.

Exit under conflict. Frustration-driven churn implements Hirschmanian exit (Hirschman 1970) under unmoderated conflict.

3.14 These are qualitative anchors, not quantitative calibration targets.

Statistical Methodology

3.15 Final-step metrics across 120 runs use non-parametric tests. Pairwise contrasts use two-sided Mann–Whitney U (Mann & Whitney 1947) with Bonferroni correction and Cohen's d (Cohen 1988). We also report 95% bootstrap intervals (10,000 resamples).

3.16 For Bridge × Trust moderation we form seed-paired effects Δ_{alone} = (Bridge Only) − (Baseline) and $\Delta_{\text{with trust}}$ = (Full Model) − (Trust Only), then apply a Wilcoxon signed-rank test to their difference, with Bonferroni correction across metrics.

Results

4.1 Our 2×2 experiments cross bridging (ON/OFF) with dyadic trust (ON/OFF). Each cell has 30 runs of 200 steps with seed $42 + i$ for run i. We compare final metrics with Mann–Whitney U tests (Bonferroni-corrected) and Cohen's d (Cohen 1988). Very large d values for churn and revenue reflect nearly separated distributions; we therefore also report absolute means.

The Polarization Paradox: Short-Term Monetization vs. Long-Term Alignment

4.2 We begin by comparing the Baseline and Bridge Only conditions, isolating the impact of algorithmic feed moderation when network trust remains static. The simulations demonstrate a tension between a platform's short-term financial retention and long-term ideological alignment—a structural deadlock we term the Polarization Paradox.

User Retention and Platform Revenue

4.3 In the unmoderated Baseline, cumulative churn reaches 39.18% (SD = 2.27%, 95% CI = $[38.39\%, 39.98\%]$) by step 200, leaving only 304.1 active users on average (SD = 11.35, 95% CI = $[300.10, 308.03]$). Frustration among the survivors rises to 0.9668 (SD = 0.1669, 95% CI = $[0.9055, 1.0239]$). This user flight depresses platform economics: final step-wise revenue falls to a mean of \$121.64 (SD = \$4.54, 95% CI = $[\$120.04, \$123.21]$).

4.4 Activating the bridging algorithm (Bridge Only condition) changes retention sharply. Cumulative churn falls to 0.38% (SD = 0.30%, 95% CI = $[0.27\%, 0.49\%]$) at step 200 (see Figure 1). The contrast is highly significant: Mann-Whitney $U = 900.0$, $p_{\text{adjusted}} < 1.5 \times 10^{-9}$, Cohen's $d = 23.97$. The platform retains 498.1 active users (SD = 1.52, 95% CI = $[497.57, 498.63]$)—nearly the full population.

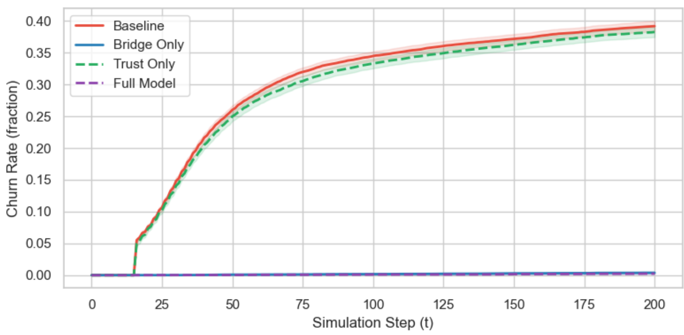

Figure 1: Cumulative user churn rate over time across the four experimental conditions ($N = 500$, 30 runs per condition). The Bridge ON conditions show near-zero user flight (0.38%), while the Bridge OFF conditions display high user attrition ($\approx 39\%$). Error ribbons indicate 95% confidence intervals.

4.5 Algorithmic shielding lowers mean frustration among active users to 0.5819 (SD = 0.0576, 95% CI = $[0.5617, 0.6024]$, Mann-Whitney $U = 871.0$, $p_{\text{adjusted}} < 3.1 \times 10^{-8}$, Cohen's $d = 3.08$). Retaining users stabilizes final step-wise revenue at \$199.24 (SD = \$0.61, 95% CI = $[\$199.03, \$199.45]$, Mann-Whitney $U = 0.0$, $p_{\text{adjusted}} < 1.5 \times 10^{-9}$, Cohen's $d = -23.97$). This approximately 64% revenue increase is driven by retention of active users rather than improved feed safety. Because the initial society is pre-polarized ($\sigma(0) \approx 0.7 > 0.6$), the stylized brand-safety multiplier remains in the unsafe regime ($M_{\text{safety}} = 0.4$) for both conditions. The financial benefit is therefore demographic: the algorithm keeps users active and preserves ad impressions even when content remains polarized.

The Paradoxical Ideological Backlash

4.6 Beneath the financial surface, filtering confrontational interactions yields a statistically significant increase in global ideological polarization. Baseline polarization σ settles at a mean of 0.9550 (SD = 0.0389, 95% CI = $[0.9409, 0.9683]$), while Bridge Only raises final-step polarization to 0.9904 (SD = 0.0042, 95% CI = $[0.9888, 0.9918]$)—a significant increase (Mann-Whitney $U = 135.0$, $p_{\text{adjusted}} < 0.0002$, Cohen's $d = -1.28$; see Figure 2).

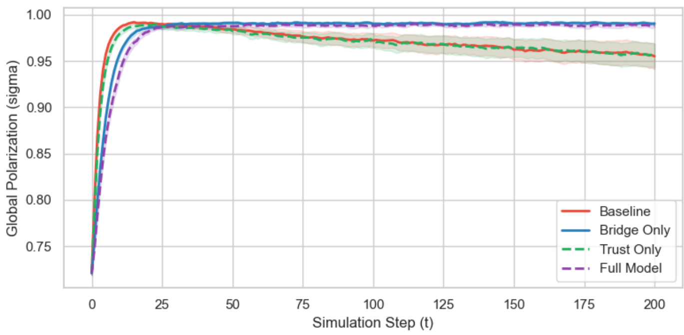

Figure 2: Global opinion polarization (σ) over time across the four experimental conditions. While the Bridging Algorithm suppresses churn in the Bridge Only condition, it paradoxically locks systemic polarization at a significantly higher level ($\sigma \approx 0.9904$) compared to the Baseline ($\sigma \approx 0.9550$), demonstrating the Polarization Paradox. Error ribbons represent 95% confidence intervals.

4.7 A structural selection effect explains this phenomenon. In the unmoderated Baseline, extreme, dogmatic agents repeatedly clash with their opponents. These interactions trigger backfires, pile up frustration, and drive agents to churn. Their permanent exit mathematically clips the tails of the opinion distribution, capping global polarization.

4.8 By contrast, the bridging algorithm isolates extreme users from frustration and keeps them active. Retained ideologues continue to pull neighbors toward the extremes, reducing the chance of consensus. By mitigating the symptoms of polarization—exit and frustration—the algorithm can preserve high systemic dispersion, locking the system in ideological deadlock.

Trust Segregation: The Relational Cost of Social Judgments

4.9 Enabling dyadic trust (Trust Only condition) triggers a dramatic structural transformation. Trust co-evolves with dialogue as a dynamic link attribute, reshaping the network's qualitative architecture.

4.10 In our Baseline, trust remains locked at a static nominal value of 0.5 across all edges, presenting a flat relational landscape where ingroup and outgroup trust are identical (yielding a segregation ratio of exactly 1.00).

4.11 Under the Trust Only condition, co-evolving opinions and trust split the network into relational camps (see Figure 3). Ingroup trust—the trust linking agents sharing the same opinion pole—climbs to a near-perfect mean of 0.9814 (SD = 0.0055, 95% CI = $[0.9794, 0.9832]$). Conversely, outgroup trust completely collapses, bottoming out at a mean of 0.0415 (SD = 0.0256, 95% CI = $[0.0331, 0.0511]$).

4.12 This relational divergence raises the Trust Segregation Ratio to a mean of 35.61 (SD = 31.50, 95% CI = $[26.29, 47.64]$). Because the ratio is heavy-tailed (maximum ≈ 175 across runs), we also report the median and interquartile range: median = 28.58, IQR = $[19.06, 40.61]$. Compared to the Baseline, the effect is large: Mann-Whitney $U = 0.0$, $p_{\text{adjusted}} < 7.3 \times 10^{-11}$, Cohen's $d = -1.55$. These figures describe the Trust Only vs. Baseline contrast (enabling trust co-evolution), not a bridging effect.

4.13 This segregation exposes the double-edged role of trust. Inside ideological camps, high ingroup trust increases the influence weight w_{ij}, buffering minor disagreements via a "friends can argue" mechanism. Across camps, the asymmetry of trust co-evolution ($\delta_-/\delta_+ = 2.5$) erodes cross-cutting ties faster than agreement can rebuild them. Combined with passive decay ($\lambda = 0.001$), outgroup trust collapses and the network segregates into two cohesive but mutually hostile relational silos.

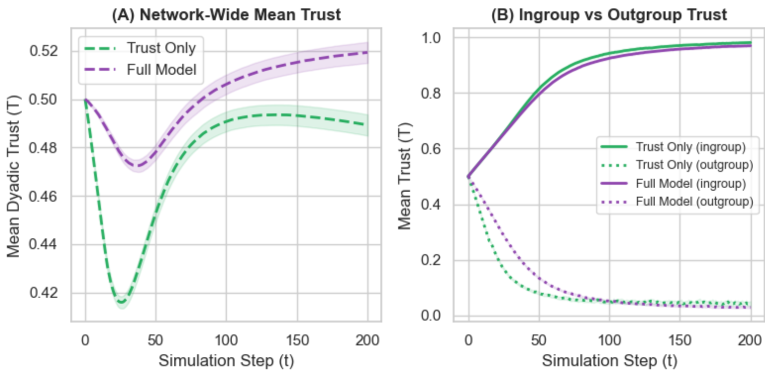

Figure 3: Interpersonal dyadic trust co-evolution dynamics. Panel A compares mean trust over time in the Trust Only and Full Model. Panel B illustrates the extreme divergence between ingroup trust (climbing toward 0.98) and outgroup trust (collapsing to near 0.03), demonstrating the emergence of severe trust segregation. Error ribbons denote 95% confidence intervals.

Moderation Analysis: The Bridge × Trust Interaction

4.14 To test whether co-evolving dyadic trust moderates the impact of algorithmic content moderation, we perform a seed-paired interaction analysis. Because runs share random seeds, we compute paired differences in final metrics:

$$\Delta_{\text{alone}} = \text{Metric(Bridge Only)} - \text{Metric(Baseline)} \tag{25}$$
$$\Delta_{\text{with trust}} = \text{Metric(Full Model)} - \text{Metric(Trust Only)} \tag{26}$$

We then apply a Wilcoxon signed-rank test to the per-seed contrast $\Delta_{\text{with trust}} - \Delta_{\text{alone}}$, which matches the paired design (see Table 2).

Table 2: Bridge × Trust Interaction Analysis (Seed-Paired Differences)

Metric	Effect Alone (Δ_{alone})	Effect w/Trust ($\Delta_{\text{with trust}}$)	Interaction Cohen's d	W-statistic	Adj. p-value
Active Users	+194.0	+190.1	−0.3508	112.5	0.1346
Avg. Frustration	−0.3849	−0.3981	−0.0739	198.0	1.0000
Churn Rate	−38.80%	−38.02%	+0.3508	116.5	0.1698
Ingroup Trust	0.0000	−0.0114	−2.6925	4.0	$< 1.4 \times 10^{-7}$
Mean Trust	0.0000	+0.0300	+4.8926	0.0	$< 1.9 \times 10^{-8}$
Outgroup Trust	0.0000	−0.0127	−0.8219	97.0	0.0434
Polarization	+0.0353	+0.0322	−0.0887	187.0	1.0000
Revenue	+$77.60	+$76.04	−0.3508	106.5	0.0954
Trust Segregation	0.0000	+1.9854	+0.0902	135.0	0.4491

Active Users, Frustration, and Churn:

4.15 Interaction effects on physical user retention (Active Users, Churn Rate, and Avg. Frustration) and financial performance (Revenue) remain non-significant after Bonferroni correction in the seed-paired analysis. The bridging algorithm's retention benefits are similar with or without trust co-evolution: it rescues approximately 190 to 194 users from churning, reduces frustration by ≈ 0.39, and preserves $\approx$ \$76 in step-wise revenue.

Outgroup Trust and Trust Segregation:

4.16 Relational metrics show a nuanced pattern. Seed-paired Wilcoxon tests find a significant Bridge × Trust interaction for Outgroup Trust ($W = 97.0$, $p_{\text{adjusted}} = 0.0434$, interaction $d = -0.82$): when trust co-evolves, the bridge effect on outgroup trust is a small mean reduction of -0.0127 (Full Model

0.0288 vs. Trust Only 0.0415). By contrast, the Trust Segregation interaction is not significant after Bonferroni correction ($W = 135.0$, $p_{\text{adjusted}} = 0.4491$, $d \approx 0.09$). Pairwise Full Model vs. Trust Only contrasts for both Outgroup Trust and Trust Segregation are also non-significant after Bonferroni correction ($p_{\text{adjusted}} = 1.000$).

4.17 We therefore treat bridging's relational side-effects as modest and method-dependent: retention benefits are robust, while claims of intensified segregation under the Full Model are not supported by the paired Wilcoxon interaction test or by pairwise condition contrasts. Mechanistically, bridging can replace rejection-zone events with Zone 2 silence (no trust rebuild) and retain agents who would otherwise churn, leaving ties exposed to passive decay ($\lambda = 0.001$); these mechanisms can shift descriptive means without producing a large, reliably detected segregation interaction.

Echo Chambers: Emergent Structural Clustering

4.18 To analyze how these cognitive and relational forces reshape the topological communication patterns of the platform, we track the Opinion-Weighted Clustering Coefficient (see Figure 4). This metric measures the local density of triadic relationships where all three connected agents share the same opinion pole—serving as a direct indicator of structural echo chambers.

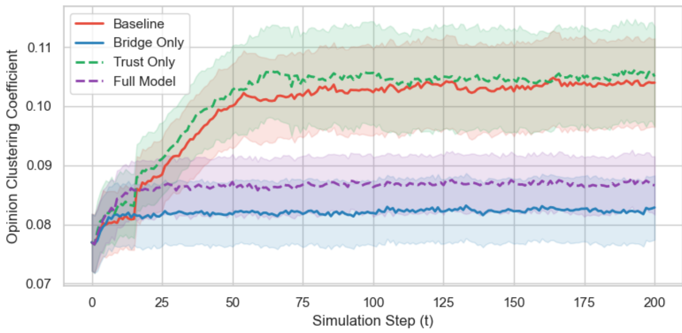

Figure 4: Emergent structural echo chambers, measured via the opinion-weighted clustering coefficient over time. The Bridging Algorithm (Bridge Only and Full Model) significantly reduces local triadic homogeneity compared to the Baseline, although global polarization remains extreme. Error ribbons show 95% confidence intervals.

4.19 In our Baseline, the final mean opinion-weighted clustering coefficient settles at 0.1040 (SD = 0.0209, 95% CI = $[0.0968, 0.1115]$). When we activate trust co-evolution alone (Trust Only), clustering remains statistically unchanged at a mean of 0.1050 (SD = 0.0237, 95% CI = $[0.0968, 0.1134]$, Mann-Whitney $U = 420.0$, $p_{\text{adjusted}} = 1.0000$, Cohen's $d = -0.05$). This static result shows that trust co-evolution alone, despite locking relationships into hostile camps, does not alter the physical clustering of opinions beyond the baseline scale-free structure.

4.20 Deploying the bridging algorithm reduces structural clustering. Under the Bridge Only condition, opinion-weighted clustering drops to a mean of 0.0829 (SD = 0.0152, 95% CI = $[0.0774, 0.0881]$)—a significant decline relative to the Baseline (Mann-Whitney $U = 721.0$, $p_{\text{adjusted}} = 0.0038$, Cohen's $d = 1.16$).

4.21 Similarly, under the Full Model (Bridge ON, Trust ON), opinion-weighted clustering settles at a mean of 0.0866 (SD = 0.0137, 95% CI = $[0.0816, 0.0914]$), also significantly below Baseline (Mann-Whitney $U = 681.0$, $p_{\text{adjusted}} = 0.0393$, Cohen's $d = 0.98$).

4.22 This shift occurs because bridging intercepts many extreme backfires and dampens local repulsion. Global polarization can remain high because dogmatic users who would otherwise exit are retained, while local triangles become slightly less homogeneous.

4.23 Lower clustering should not be read as systemic resolution. Agents may remain connected in triangles while interpersonal trust stays partitioned and many cross-cutting channels are inactive.

Mechanism Checks

4.24 To isolate what produces the paradox, we ran paired ablations (20 seeds; Figure 5). Relative to an unmoderated reference (mean churn 38.66%, polarization 0.9579), standard intercept bridging (M0) cuts churn to 0.39% while raising polarization to 0.9901. A placebo that heals frustration without intercepting opinion change (M1) lowers churn only to 5.82% and leaves polarization high (0.9924): retention help without opinion intercept is weaker than full bridging. Removing the frustration-healing bonus while keeping intercept (M3) raises churn to 20.53%, so healing contributes to retention but is not the whole story. Dropping rejection events entirely (M2) matches intercept on these aggregate KPIs under our defaults. Disabling passive trust decay in the Full Model (M4) still yields near-zero churn and high polarization. Setting the brand-safety multiplier to 1.0 (M5) leaves polarization and churn unchanged but raises revenue to the full active-user scale, confirming that the cliff mainly scales revenue rather than the opinion paradox.

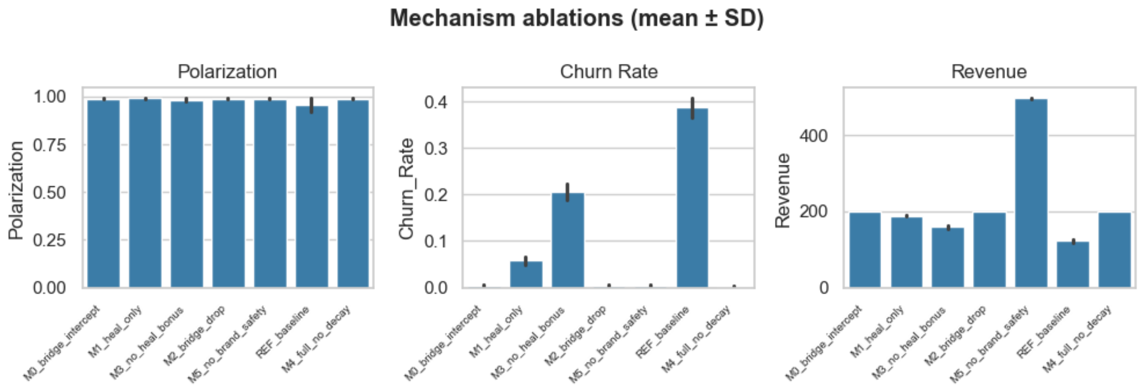

Figure 5: Mechanism ablations (20 seeds): mean Polarization, Churn Rate, and Revenue by condition. Error bars show standard deviations.

Baseline Model Comparison

4.25 On the same BA networks and seeds, simple Hegselmann–Krause bounded confidence yields lower final polarization (mean 0.6991) and no churn pathway. A lightweight biased-assimilation update without Social Judgment zones or exit also stays near-extreme (mean polarization 0.9900) with zero churn. Dynamic-BABE Baseline is distinctive in jointly producing high polarization and substantial exit (churn 38.66%). Bridging therefore acts on a model that already links conflict to leaving—a coupling the simpler baselines omit (Figure 6).

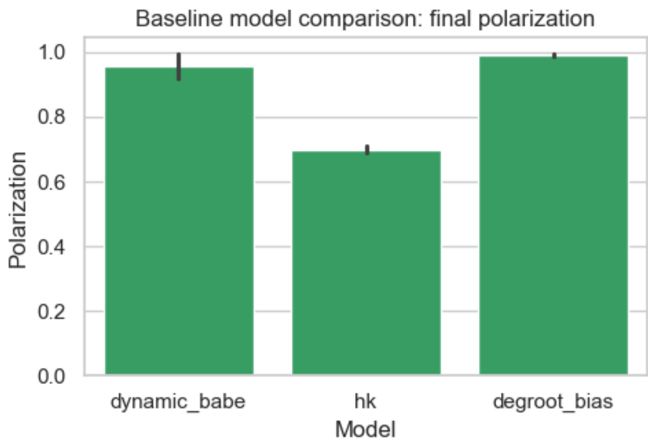

Figure 6: Final polarization under Dynamic-BABE Baseline, Hegselmann–Krause, and biased DeGroot-style updates (20 seeds).

Topology Robustness

4.26 Repeating Bridge OFF/ON on Watts–Strogatz graphs ($k = 6$, $p = 0.1$; 15 seeds) preserves the qualitative paradox: bridging cuts churn from 36.04% to 0.17% while polarization rises from 0.9730 to 0.9896, comparable in direction to Barabási–Albert under the same seeds (Figure 7).

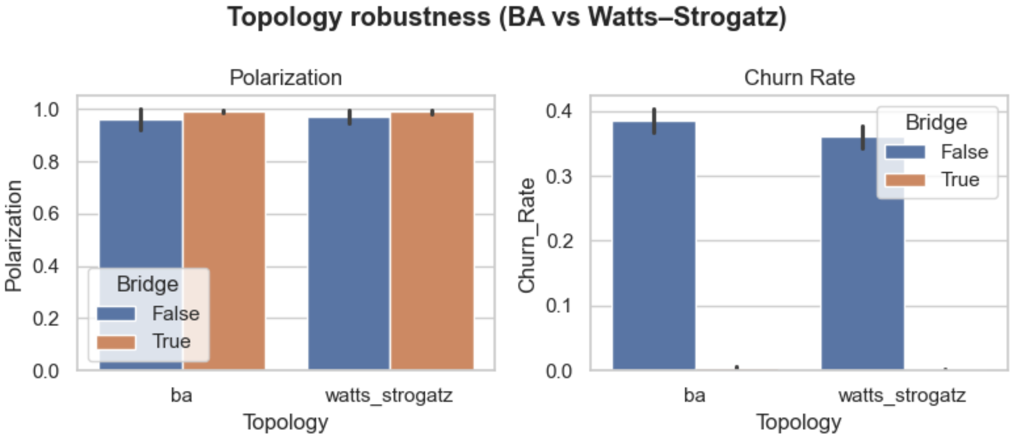

Figure 7: Bridge OFF/ON contrasts on Barabási–Albert vs. Watts–Strogatz topologies (15 seeds).

Discussion

5.1 We interpret the Polarization Paradox with Exit–Voice–Loyalty, discuss trust, state conditional implications, summarise robustness (including mechanism and topology checks), and list limitations.

The Polarization Paradox: Exit, Voice, and Loyalty

5.2 Hirschman (1970) distinguishes Exit (leave) from Voice (stay and try to repair). In the Baseline, rejection-zone contact raises frustration and many agents exit. Exit is costly for the platform, but it also removes some extreme agents from the active population.

5.3 Bridging intercepts many backfires and reduces exit. Retention and revenue improve. The same intervention often replaces rejection with Zone 2 silence rather than constructive engagement, so it does not build Voice. Closing exit without opening repair can leave extreme agents active, sustaining high polarization while churn falls.

Dyadic Trust

5.4 Trust is double-edged. Agreement raises ingroup trust (mean 0.9814 in Trust Only); conflict and asymmetry ($\delta_-/\delta_+ = 2.5$) following Slovic (1993) collapse outgroup trust (mean 0.0415). With passive decay, trust segregates (mean ratio 35.61; median 28.58). These patterns come mainly from enabling trust co-evolution, not from the bridging contrast that defines the paradox.

Conditional Implications

5.5 Curation systems often optimise engagement, retention, and brand safety. Our results suggest those objectives need not track cohesion. If designers also care about cross-camp relational capital, suppression alone is an incomplete instrument. That statement is a conditional implication from a theoretical laboratory, not a prescription for any named platform.

Robustness

5.6 OFAT sweeps over β_{mean}, τ_{assim}, T_c, K, γ_T, E, and m_e (five replications per cell; $N = 400$) preserve the qualitative paradox at interior points: bridging cuts churn and typically raises final polarization.

Exceptions are identifiable boundaries (very low β; $E = 0$; near-perfect $E = 1$). See `sensitivity.py` and `output/sensitivity_results.csv`.

5.7 Additional checks (see Results) show that (i) placebo retention without opinion intercept does not reproduce the paradox in the same way as intercept bridging, (ii) Dynamic-BABE differs from simple HK / biased-DeGroot baselines in linking suppression to exit, and (iii) the Bridge OFF/ON contrast remains directionally similar on Watts–Strogatz as on Barabási–Albert graphs under our defaults.

Limitations and Future Work

1. Symmetric trust: We assume $T_{ij} = T_{ji}$.
2. Static undirected topology: BA/WS graphs are undirected and fixed; real follower graphs are directed and rewiring.
3. Low-dimensional issues: $K = 2$.
4. Pole edge case: $\text{sign}(\overline{O}_i) = 0$ is a distinct pole (rare under bipolar init).
5. No field fit: Parameters are not estimated from a named platform dataset.
6. Stylized bridge and revenue: Abstractions for theory, not audited production systems.

5.8 Despite these limits, the model is a transparent testbed for how suppression and fragile trust jointly shape polarization and retention.

Conclusion

6.1 Dynamic-BABE links opinion dynamics, dyadic trust, and a stylized bridging filter. Under our defaults, bridging sharply improves retention and step-wise revenue while raising final polarization by keeping extreme agents who would otherwise exit. Trust co-evolution separately produces trust segregation. Seed-paired Bridge × Trust analyses suggest retention benefits are robust to trust, whereas relational moderation should be read cautiously.

6.2 The practical implication is modest: retention and cohesion need not move together when disagreement is suppressed and trust is fragile. Designs that only minimise visible conflict may miss relational outcomes that matter for long-run tolerance. Escaping the paradox, if that is the goal, requires contact conditions under which trust can form—not only filters that prevent backfire.

Code and Data Availability

6.3 The Dynamic-BABE implementation, batch scripts, analysis pipeline, and figure generators are publicly available at `https://github.com/TheCoolSam/BABEModel`. We intend to deposit a reviewed release on the CoMSES Net Computational Model Library so that a permanent handle can accompany the published article.

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Notes

[1]CoMSES upload to be completed by the authors prior to publication.

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